ISDA FRTB-SA: RISK DATA STANDARDS

Version 1.8: 26 January 2024

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Track changes

Version	Date	Changes description
1.1	26/03/2020	Extensive rewriting
1.1b	02/02/2022	Adding RRAO and other minor changes
1.2	21/02/2023	Made changes to align content with unit test version 3.2.1. including: • Decommissioning of prior curvature variants 1a and 1b • Changes to drc ns variants
1.3	08/03/2023	Fixed broken footnote references, fixed GIRR Delta bucket and qualifier description error
1.4	22/03/2023	Commodity Bucket datatype is updated to string to support 3a, 3b buckets under UK-PRA jurisdiction
1.5	25/07/2023	Added section 8 on ISDA tools ecosystem built around CRIF and relevant configuration flags. Other minor edits.
1.6	23/08/2023	Made changes to align with the latest release of the unit-test v3.4.0 and the introduction of the US-NPR jurisdiction. Also includes changes related to repurposing Variant 2 of DRC_SNC and reconfiguring the representation of DRC_SC inputs. Finally we introduce a new section 8, where we give detailed description of the various input fields and their data types per jurisdiction and risk type.
1.7	19/12/2023	Minor changes: • fixed typo in Table 18 from “Variant 1” to “Variant 2” • Removed “Bucket” row from Table 22 as unused. • fixed typo in Table 42 Label2, US-NPR from “GSE_NOT_GURANTEED” to “GSE_NOT_GUARANTEED”
1.8	26/01/2024	Minor changes: • Updated the description of CSR_NS_INDX_BUCKET_NAME_CORRELATION to be in line with the changes introduced in unit test version 3.4.1 where based on UK-PRA’s near final rules the ρ^{name} correlation is fixed to 80%. In other words, the particular control flag no longer applies to the UK-PRA jurisdiction. • As above but for CRR2_CSR_NS_INDX_RATING_CORR.

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1 Introduction

Basel Committee has revised the market risk framework for calculation of minimal capital requirements for market risk, with the latest standard published on December 2019, see [0]. This revision called Fundamental Review of the Trading Book (FRTB) produced a new and more risk sensitive Standardised approach (SA). As a response ISDA launched an industry wide benchmarking exercise to help measure and hopefully decrease the degree of variability in the implementation of the FRTB-SA model. The process requires exchange of risk data for a number of benchmark portfolios, between participating institutions and ISDA. Therefore, a structured and standardised format to transmit such data was developed.

This document sets complete standards and describes the data format used for data exchange and risk calculations, giving additional descriptions and clarifications of the various risk factors and other inputs, as well as worked examples.

As part of that, the document also proposes a common standard for interchange of FRTB SA risk between participants and others.

2 Common Risk Interchange Format (CRIF)

There is a strong need for a common way of describing risk on portfolios and trades. This is required by market participants for a number of use cases of which a non-exhaustive list can be found below.

- Third party vendors which aim to provide FRTB reconciliation or management services.
- ISDA member firms for back-testing, benchmarking and other validation processes.
- New firms can be benefited by having a swift and precise on-boarding phase related to the FRTB-SA regulatory capital methodology.

The proposal is for a simple, robust format, which can be read by automated processes as well as being capable of manual inspection. The format is specified for the current set of risk factors, but can also easily be extended to handle new risk factors in future.

The objective is to have a format which is capable of achieving two different objectives – both risk reconciliation and the basis for regulatory capital calculation. Given risk data in this format, it should be straightforward and mechanical to calculate the regulatory capital of a portfolio, using only additional information from the FRTB Methodology document, [0].

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3 Format overview

The format can be implemented as an Excel file or in any other column based file extension like csv or txt, with defined data fields arranged in columns. Elements can be either textual (string), date or numerical and type can be context-dependent.

The first row of the Excel file contains the header with the names of the columns.

Each additional row contains a single risk factor for a trade or portfolio.

Below we provide a quick overview of each column. For more details see "4. Fields description".

1. **Portfolio ID** - string describing portfolio ID.
2. **Trade ID** – string describing the trade ID.
3. **Variant** - string describing the variant of the input format. This string is only relevant for Vega, Curvature and DRC[1].
4. **Sensitivity ID -** string describing the risk factor ID.
5. **RiskType** – string describing the risk factor type, such as general interest-rate risk (GIRR), non-securitisations credit risk, etc. Specific examples include GIRR_DELTA (interest-rate delta), FX_DELTA (FX delta risk), and others[2].
6. **Qualifier** – string description of the particular risk factor. Depends on the RiskType. For example for GIRR, the Qualifier denotes the ISO currency code of and issuer name in case of equities
7. **Bucket** – Denotes the FRTB-SA bucket as per the regulations. For GIRR, FX denotes the integer code relating to liquidity category: 1- Regular, 2- Liquid of currency name of the Qualifier we are trying to model. Exact meaning, interpretation depends on RiskType[3].
8. **Label1** – additional qualifier, typically describing the tenor of an Option (for vega RiskTypes) or the shift necessary to calculate CVR (for curvature RiskTypes).
9. **Label2** – additional qualifier used to further specify the type of risk factor within the row. Meaning depends on the RiskType.
10. **Amount** – amount of the risk, in units of a given currency. Number format. Can have different meaning based on RiskType or Variant. For example, in the case of RRAO it is the Notional of the position.
11. **AmountCurrency** – currency of the "Amount" field. Should be a standard three-letter ISO currency code.
12. **AmountUSD** – amount of the risk, expressed in USD. (This has the additional benefit of implying the effective FX rate used by the data-producer to convert between the amount currency and USD.)

Additional columns:

13. **Label3** – used to input volatility for vega risk for all risk types or notional for calculating DRC for DRC_NS risk type for 'Variant 1'. For all other positions it is blank.
14. **EndDate** – is a maturity date for credit instruments and is used only for DRC. The date must be entered in "YYYY-MM-DD" format, according to the ISO 8601 standard. For other non-relevant positions it must remain blank.

[1] See "5. Variants description" for more information and available options.
[2] For the full list see Table 1, Table 2 and Table 3.
[3] For a detailed list see "**Error! Reference source not found.**".

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15. **CreditQuality** – is either an issuer rating class for credit instruments or a direct risk-weight input. Depending on the jurisdiction it can follow a standard Moody's rating nomenclature, e.g. "AAA", or Credit Quality Step (CQS) nomenclature etc. For other irrelevant positions it must remain blank.
16. **LongShortInd** – Long or Short indicator. It is used only for DRC calculation for credit instruments. It must take either "L", "S" or blank for irrelevant positions.
17. **CoveredBondInd** – takes values, "Y" for yes, "N" for no and blank if not relevant. The field applies only to calculation of DRC for credit instruments which are non-securitisations.
18. **TrancheThickness** – is a number between 0 and 1 and is used for DRC calculation for Securitisation non-CTP positions under Variant 1. It is used for the calculation of the relevant risk-weights based on the SEC-ERBA methodology.

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4 Fields description

Where dates are used in CRIF files, they should be formatted according to the ISO 8601 standard, which can be described as "YYYY-MM-DD", so that 1 September 2016 would be "2016-09-01". This format is unambiguous, and has useful sorting properties.

Below we give a detailed description of the various fields:

RiskType: String denoting the FRTB-SA risk class and risk combination for the Sensitivity Based Method (SBM) related calculations. Alternative options provided for the Default Risk Charge (DRC) specification.

SBM risk class	Delta	Vega	CURVATURE
General Interest Rate Risk (GIRR)	GIRR_DELTA	GIRR_VEGA	GIRR_CURV
Credit spread risk (CSR): non-securitisations	CSR_NS_DELTA	CSR_NS_VEGA	CSR_NS_CURV
CSR: securitisations (non-correlation trading portfolio, or non-CTP)	CSR_SNC_DELTA	CSR_SNC_VEGA	CSR_SNC_CURV
CSR: securitisations (correlation trading portfolio, or CTP)	CSR_SC_DELTA	CSR_SC_VEGA	CSR_SC_CURV
Equity risk	EQ_DELTA	EQ_VEGA	EQ_CURV
Commodity risk	COMM_DELTA	COMM_VEGA	COMM_CURV
Foreign exchange (FX) risk	FX_DELTA	FX_VEGA	FX_CURV

Table 1: RiskType options for SBM calculation

DRC risk class	All Variants
Non-securitisation portfolios	DRC_NS
Securitisation portfolio (non-correlation trading portfolio, or non-CTP)	DRC_SNC
Securitisation (correlation trading portfolio, or CTP)	DRC_SC

Table 2: RiskType options for DRC calculation

Residual Risk Add-On risk class	Risk Type
Exotic	RRAO_1_PERCENT
Other residual risk	RRAO_01_PERCENT

Table 3: Risk Type options for RRAO

Qualifier: String denoting the risk factor at an "in-bucket" level.

SBM risk class	Meaning
GIRR_DELTA ,GIRR_CURV, FX_DELTA, FX_CURV	ISO Currency code, e.g., USD, EUR
CSR non-sec	Issuer or Index name/ID
CSR sec nCTP	Issuer, Tranche or Index name/ID
CSR sec CTP	Issuer, Tranche or Index name/ID
Equity	Issuer, Index name/ID
Commodity	Commodity name/ID

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FX_VEGA	ISO Currency code of FX pair, e.g., GBPUSD, EURHUF
DRC non-sec	Issuer name/ID
DRC sec nCTP	Issuer/Tranche ID
DRC sec CTP	Series of the index or "Bespoke portfolio"

Table 4: Qualifier field guidance

Bucket: This field typically denotes the Bucket the particular sensitivity belongs to but its actual meaning depends on the *RiskType* it is used with:

RiskType	Meaning	Type and example
GIRR_DELTA ,GIRR_CURV, FX_DELTA, FX_CURV	Integer code relating to liquidity category of respective currencies:	1- Regular, 2- Liquid
GIRR_VEGA, FX_VEGA	Null	
CSR_NS_DELTA, CSR_NS_VEGA, CSR_NS_CURV	Credit Spread – Non-Securitisation bucket	Integer e.g. 1
CSR_SC_DELTA, CSR_SC_VEGA, CSR_SC_CURV	Credit Spread – Securitisation CTP bucket	Integer e.g. 1
CSR_SNC_DELTA, CSR_SNC_VEGA, CSR_SNC_CURV	Credit Spread – Securitisation non-CTP bucket	Integer e.g. 1
EQ_DELTA, EQ_VEGA, EQ_CURV	Equity bucket	Integer e.g. 1
COMM_DELTA, COMM_VEGA, COMM_CURV	Commodity bucket	String e.g. 3, 3a
DRC_NS	DRC Non-Securitisation buckets	String denoting the DRC_NS bucket for example "CORPORATES" or "SOVEREIGNS".
DRC_SNC	DRC Securitisation non-CTP buckets	String denoting the DRC_SNC bucket for example "CORPORATES" or "RMBS_EUROPE"[4].
DRC_SC	DRC Securitisation CTP buckets	String denoting the name of the issuer/index.[5]

Table 5: Bucket field guidance

Label1: This field has different meanings depending on the *RiskType* it is used with:

RiskType	Meaning	Type and example
GIRR_DELTA,	Tenor (year fraction)	Number e.g. 5
GIRR_VEGA	Option tenor	Number e.g. 5
GIRR_CURV	Rate shift for calculating the CVR	Number e.g. 0.0001
CSR_NS_DELTA, CSR_SNC_DELTA, CSR_SC_DELTA	Tenor (year fraction)	Number e.g. 5

[4] For DRC Securitisations non-CTP the two main buckets are "CORPORATES" and other. Other is defined along two dimensions – asset class and region. For more details see MAR22.31.

[5] A non-exhaustive list of indices include: "CDX North America IG", "iTraxx Europe IG", "CDX HY", "iTraxx XO" etc.

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CSR_NS_VEGA, CSR_SNC_VEGA, CSR_SC_VEGA	Option tenor	Number e.g. 5
CSR_NS_CURV, CSR_SNC_CURV, CSR_SC_CURV	Rate shift for calculating CVR	Number e.g. 0.0001
EQ_DELTA, FX_DELTA	-	<blank>
EQ_VEGA, FX_VEGA	Option tenor	Number e.g. 5
EQ_CURV, FX_CURV	Shift for calculating CVR	Number e.g. 0.01
COMM_DELTA	Tenor (year fraction)	Number e.g. 1
COMM_VEGA	Option tenor	Number e.g. 1
COMM_CURV	Shift for calculating CVR	Number e.g. 0.01
DRC_NS, DRC_SNC	-	<blank>
DRC_SC	Names in the tranche.	Range. For example for ITRAXX IG it could be 1, 2, 3, 4 – 125.

Table 6: Label1 field guidance

Label2: This field has different meanings depending on the *RiskType* it is used with:

RiskType	Meaning	Type and example
GIRR_DELTA,	Name of the curve used.	String e.g. "Libor3m"
CSR_NS_DELTA, CSR_SC_DELTA, CSR_SNC_DELTA	Relevant issuer credit spread curve	"BOND" or "CDS"
GIRR_VEGA	Underlying maturity tenor	Number e.g. 5
EQ_DELTA	Denotes whether the sensitivity is against a "SPOT" equity risk factor or against an equity "REPO" rate.	"SPOT" or "REPO"
COMM_DELTA	Commodity delivery location	String e.g. "LONDON"
DRC_NS	Instrument LGD category	e.g. "EQUITY", "SENIOR", "NON-SENIOR". Depends on the jurisdiction[6].
DRC_SNC	Variant 1 (SEC-ERBA): Instrument seniority	"SENIOR", "NON-SENIOR"
	Variant 2: Not applicable	<blank>
DRC_SC	Characterises the type of instrument that gives rise to the exposure.	One of: {TRANCHED_INDEX, NON_TRANCHED_INDEX, BESPOKE_EXPOSURE, SINGLE_NAME_HEDGE}
Otherwise	Not applicable	<blank>

Table 7: Label2 field guidance

Label3: This field has different meanings depending on the *RiskType* it is used with:

- GIRR_VEGA, CSR_NS_VEGA, CSR_SNC_VEGA, CSR_SC_VEGA, EQ_VEGA, COMM_VEGA, FX_VEGA. When Variant2 is used[7], Label3 is used to denote the implied volatility of the option.

[6] For details see Table 42: Field Guidance - DRC_NS.
[7] See "5. Variants description" for more information available options related to Variants.

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- DRC_NS: Notional for Variant 1. Negative amount signifies a short (in the DRC sense) position. Not required for Variants 2 and 3.
- Otherwise: <blank>.

EndDate: Maturity date or option expiry in ISO-Standard "YYYY-MM-DD". Applies to below *RiskTypes* (otherwise <blank>):

- DRC_NS (except from Variant 3)
- DRC_SNC
- DRC_SC

It is used to scale DRC positions according to maturity assuming an ACT/365 day-count convention.

CreditQuality: Issuer's credit quality for example "AAA" for Basel ruleset and "CQS1" for CRR ruleset. Applies to *RiskTypes* (otherwise <blank>). In some cases we use the risk-weight as a direct input for example in DRC_SNC Variant 2 and DRC_SC Variant 1 (Banking Book approach):

- CSR_NS_DELTA
- CSR_NS_CURV
- DRC_NS,
- DRC_SNC,
- DRC_SC

LongShortInd: "L" (long) or "S" (short) indicator. Applies to DRC *RiskTypes,* DRC_NS, DRC_NS, DRC_SNC, DRC_SNC, DRC_SC, DRC_SC. Blank otherwise.

CoveredBondInd: Boolean. "Y" or <blank>. Applies only to DRC_NS risk type.

TrancheThickness: Number between 0 and 1 describing the tranche thickness. Applies to the DRC_SNC Variant 1 (SEC-ERBA).

Amount: This field has different meanings depending on the *RiskType* it is used with. It is always a float. For more information see Table 8: Input Variants.

AmountCurrency: The currency of the value under the Amount column

AmountUSD: Amount equivalent expressed in USD.

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5 Variants description

Since the infrastructure developed for calculation of standardised capital for market risk varies among banks, various input variants are permissible. These variants are equivalent ways to describe the inputs that correspond to a specific risk calculation and they constitute valid data formats that are described in the Table below. In certain cases, Variants are used to model different calculation methods permissible by the rules.

Area	Variant Identifiers
Delta Calculation	One variant – sensitivities provided as input
Vega Calculation	Variant 1 uses volatility weighted vega Variant 2 uses vega and the implied volatility as separate inputs
Curvature	Variant 1a is comprised of two rows. Each row under the Amount column contains the CVR_k^+ and CVR_k^- as inputs.
	Variant 2a uses three rows. It implements the CVR calculation as per the regulations. The first row contains the sensitivity s_{ik} under the Amount and the corresponding shift that was used to derive this sensitivity. The second row contains the value differential $V_i\left(x_k^{RW(Curvature)^+}\right) - V_i(x_k)$ of the asset after the curvature shock $RW_k^{Curvature}$ is applied for the CVR_k^+ under the Amount column. The $RW_k^{Curvature}$ is specified under the Label1. The thrid row contains the value differential $V_i\left(x_k^{RW(Curvature)^-}\right) - V_i(x_k)$ of the asset after the curvature shock $RW_k^{Curvature}$ is applied for the CVR_k^+ under the Amount column. The $RW_k^{Curvature}$ is specified under the Label1.
DRC_NS	Variant 1 uses position attributes (such as notional and market value) to calculate JTD.
	Variant 2 uses the Gross JTD amount (unscaled) instead of Notional.
	Variant 3 uses the Gross JTD amount scaled by maturity
DRC_SNC	Variant 1 uses the SEC-ERBA methodology to calculate the Risk-Weights for DRC_SNC. Since SEC-ERBA is not permitted by the US-NPR rule-set this variant cannot be used with US-NPR. However, it is valid for other jurisdictions.
	Variant 2 takes the Risk-Weights under the CreditQuality column as a direct input. It is used to source Risk-Weights irrespective of the feeder model used to produce them. In that way we separate the feeder model implementation to that of FRTB-SA. This variant can be used by all jurisdictions. For US-NPR it can be used to implement both the Expanded and Standardised approach calculations.
DRC_SC	Variant1 implements the Banking Book approach i.e. the Risk-Weights in the CreditQuality column are taken as an input but the user has used the relevant banking book model to calculate them.
	Variant2 implements the decompositions approach. It is assumed that once decomposed, the risk-weight for all the relevant exposures can be

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	sourced using the DRC_NS rules and the credit rating found under the CreditQuality column.
RRAO Calculation	One variant – Gross notional amount provided as input

Table 8: Input Variants

6 Variants examples

6.1 Vega – Variants 1 & 2

The "Amount" for "Variant 1" is the "vega risk sensitivity to a given risk factor"[8]. This sensitivity s_k is an input to the model and according to the regulations it is calculated as such:

$$s_k = vega \times implied\ volatility$$

Therefore, in the example from the table below, "Variant 1" directly inputs $s_k = -100$ into the model.

However, some banks prefer to input instead the $vega$ and $mplied\ volatility$ separately and leave the model itself to calculate s_k. Such banks will use "Variant 2" where the "Amount" denotes the $vega$ and "Label3" denotes the $implied\ volatility$ and therefore $s_k = -100 = -400\ \times 0.25$.

Variant	Sensitivity ID	RiskType	Qualifier	Bucket	Label1	Label2	Amount	Label3
Variant 1	S_EQV_a1	EQ_VEGA	ISSUER A	1	0.5		- 100.00	
Variant 2	S_EQV_a1	EQ_VEGA	ISSUER A	1	0.5		- 400.00	0.25

Table 9: Vega Variants

6.2 Curvature – Variants 1a & 2a

As described in MAR21.5 the inputs for the calculation of FRTB-SA Curvature metrics is a set of values called CVR^+ and CVR^-. Roughly speaking these correspond to the change in the value of an instrument when the underlying risk factor is bumped upwards (CVR^+) or downwards (CVR^-), minus the impact of the first order movement (delta impact). For an exact specification see MAR21.5.

We start the description of the curvature Variants by the simplest one.

1. **Variant 1a (Full revaluation – delta stripped)**[9]

   Under this Variant the values under "Amount" represent the CVR^+ and CVR^- respectively for a change of ±0.55 of the underlying risk factor (see under "Label1"). Every "Sensitivity ID" needs two rows as entries.

Variant	Sensitivity ID	RiskType	Qualifier	Bucket	Label1	Label2	Amount
Variant 1a	S_EQC_a1	EQ_CURV	ISSUER A	1	0.55		-37,820.00
Variant 1a	S_EQC_a1	EQ_CURV	ISSUER A	1	-0.55		39,157.50

Table 10: Curvature Variant 1a

2. **Variant 2a (Full revaluation)**

[8] For example see MAR21.25.
[9] This is the previous Variant 2b but with the actual CVR values (previously it was $-CVR$).

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This Variant is similar to "Variant 2b" but the CVR values in the last two rows are not "delta-stripped" i.e. the impact of delta in the calculation of CVRs has not been removed yet. The delta impact is represented in the first row under "Amount" per 0.01% change of the risk factor ("Label1"). Based on this information we can recover the CVRs from "Variant1a". This operation happens behind the scenes, within the FRTB-SA engine. For example:

$$CVR_{2b}^{+} = 37{,}820 = -(39{,}195 - 25 * \frac{0.55}{0.01})$$

Variant	Sensitivity ID	RiskType	Qualifier	Bucket	Label1	Label2	Amount
Variant 2a	S_EQC_a1	EQ_CURV	ISSUER A	1	0.01		25.00
Variant 2a	S_EQC_a1	EQ_CURV	ISSUER A	1	0.55		39,195.00
Variant 2a	S_EQC_a1	EQ_CURV	ISSUER A	1	-0.55		-40,532.50

Table 11: Curvature Variant 2a

6.3 DRC – Variants 1 & 2

6.3.1 DRC_NS

1. Variant 1 (Marked to Market)

This Variant is defined by using the values under "Amount" as the Market Value (Marked-to-Market) of the instrument, while "Label3" is the associate Notional. The Gross Jump-To-Default (JTD) can be calculated by the engine using the regulatorily prescribed formulas.

Variant	Sensitivity ID	RiskType	Qualifier	Bucket	Label1	Label2	Amount	Label3
Variant 1	S_DNS_a1	DRC_NS	ISSUER A	CORPORATES		SENIOR	1,200,000	1,000,000

Table 12: DRC_NS Variant 1 (M2M)

2. Variant 2 (JTD)

For this Variant, the value under "Amount" is the actual Jump-To-Default (JTD) value associated with the seniority class of the exposure, specified under columns Label2. For example for a "SENIOR" exposure the model will apply an recovery rate of 25% according to the regulations. The value under Amount is "maturity unscaled" and is scaled by the model according to the maturity of the exposure (defined under column "EndDate").

Variant	Sensitivity ID	RiskType	Qualifier	Bucket	Label1	Label2	Amount	Label3	EndDate
Variant 2	S_DNS_a1	DRC_NS	ISSUER A	CORPORATES		SENIOR	950,000		2024-08-16

Table 13: DRC_NS Variant 2 (JTD)

3. Variant 3 (Maturity Scaled JTD)

This is the same as Variant 2 above but this time the value under Amount is already maturity scaled by the user. There is no need to provide the maturity under the EndDate column.

Variant	Sensitivity ID	RiskType	Qualifier	Bucket	Label1	Label2	Amount
Variant 3	S_DNS_a1	DRC_NS	ISSUER A	CORPORATES		SENIOR	950,000

Table 14: DRC_NS Variant 2 (JTD)

6.3.2 DRC_SNC

1. Variant 1 (SEC-ERBA)

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This Variant leverages the credit rating found under the CreditQuality column, and uses the SEC-ERBA methodology to calculate the risk-weights needed to calculate the DRC capital. The value under the "Amount" column is the JTD which according to the rules (see MAR22.27) is the same as the market value of the position. TrancheThickness also plays a role to scale the risk-weight according to the SEC-ERBA methodology. Finally, Label2 entries are used to define a SENIOR or NON-SENIOR position.

Since, SEC-ERBA is not supported by the US-NPR rules, Variant 1 should not be used with it. It can be used with all other jurisdictions.

Variant	Sensitivity ID	RiskType	Qualifier	Bucket	Label1	Label2	Amount	CreditQuality	TrancheThickness
Variant 1	S_DNC_a1	DRC_SNC_M2M	ISSUER A	CORPORATES		SENIOR	1,200,000	AAA	0.5

Table 15: DRC_SNC Variant 1

2. Variant 2 (risk-weight as an input)

Similarly to Variant 1, the value under the "Amount" column is the JTD which according to the rules (see MAR22.27) is the same as the market value of the position. However, unlike Variant 1, we do not make any assumption about the feeder model that is used to calculate the risk-weights. The user inputs the risk-weights directly under the CreditQuality column. The risk-weight must be input as a float number and not as a percentage[10]. Since SEC-ERBA is not used, Label2 and TrancheThickness columns can be left empty. This Variant is applicable to all jurisdictions and can be used to model the Expanded and Standardised calculations under the US-NPR if required.

Variant	Sensitivity ID	Risk Type	Qualifier	Bucket	Amount	EndDate	CreditQuality	LongShortInd
Variant 2	S_V2_DNC_a1	DRC_SNC	ISIN A	CORPORATES	(1,200,000.00)	2019-08-15	0.012000	S
Variant 2	S_V2_DNC_a2	DRC_SNC	ISIN A	CORPORATES	1,700,000.00	2019-08-15	0.012000	L

Table 16: DRC_SNC Variant 2

6.3.3 DRC_SC

1. Variant 1 (Banking Book approach)

In this Variant each row reports the JTD number under the "Amount" column. Under the CreditQuality column the user can input the risk-weights they have calculated using the relevant Banking-Book model. Label2 entries[11] are used to denote the type of exposure being modelled. The Bucket is used to denote the relevant DRC_SC bucket (usually the index the tranche belongs to) and Qualifier the tranche's series.

Variant	Sensitivity ID	RiskType	Qualifier	Bucket	Label1	Label2	Amount	EndDate	CreditQuality
Variant 1	S_DCB_a10	DRC_SC	Series 18	ITRAXX IG	1	TRANCHED_INDEX	500,000	2019-08-15	0.012000

Table 17: DRC_SC Variant 1 (Banking Book approach)

[10] 0.01 is a good entry but 1% should be avoided.

[11] Label2 can take one of the following values: {BESPOKE_EXPOSURE, TRANCHED_INDEX, NON_TRANCHED_INDEX, SINGLE_NAME_HEDGE}

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2. Variant 2 (Decomposition approach)

Variant 2 is similar to Variant 1 but in this case it is assumed that the tranched indices have been decomposed to single names according to the relevant regulatorily described rules. In this case, the risk-weight determination can be done using the credit rating under the CreditQuality column and the relevant DRC_NS rules.

Variant	Sensitivity ID	RiskType	Qualifier	Bucket	Label1	Label2	Amount	EndDate	CreditQuality
Variant 2	S_DCB_a10	DRC_SC	Series 18	ITRAXX IG	1	TRANCHED_INDEX	500,000	2019-08-15	IG

Table 18: DRC_SC Variant 2 (Decomposition approach)

7 RRAO example

RRAO is specified in CRIF by providing the positions Notional under the Amount and AmountUSD columns. The correct risk weight is applied based on the entry under RiskType. “RRAO_1_PERCENT” is used to model positions with exotic underlyings (1% risk weight) while “RRAO_01_PERCENT” is used for instruments with other residual risks (0.1% risk weight).

Variant	Portfolio ID	Trade ID	Sensitivity ID	RiskType	Amount	AmountCurrency
	10	1	S_RRAO_a1	RRAO_1_PERCENT	1,000,000	USD
	10	2	S_RRAO_a2	RRAO_01_PERCENT	1,000,000	USD

Table 19: RRAO example

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8 Fields guidance per risk type

The following session provides guidance for each CRIF field, depending on Risk Type and Jurisdiction[12] (rule-set). The table below presents some fields that pertain to all Risk Types, namely Portfolio ID and Trade ID.

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Portfolio ID	<string>	<string>	<string>	<string>	Name of the portfolio
Trade ID	<string>	<string>	<string>	<string>	Name of the trade
AmountCurrency	<string>	<string>	<string>	<string>	Currency of sensitivity
AmountUSD	<float>	<float>	<float>	<float>	Equivalent of Amount column but in USD terms

Table 20: Field Guidance - General Fields

The following subsections will examine in more detail each individual Risk Type and the corresponding Field types.

8.1 GIRR_DELTA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<empty>	<empty>	<empty>	<empty>	Not used for Delta risk types
Sensitivity ID	<string> e.g. "S_IRD_xx"	<string> e.g. "S_IRD_xx"	<string> e.g. "S_IRD_xx"	<string> e.g. "S_IRD_xx"	Name of the sensitivity
RiskType	<string> "GIRR_DELTA"	<string> "GIRR_DELTA"	<string> "GIRR_DELTA"	<string> "GIRR_DELTA"	Name of Risk Type
Qualifier	<string> e.g. "USD", "EUR" of currency	<string> e.g. "USD", "EUR" of currency	<string> e.g. "USD", "EUR" of currency	<string> e.g. "USD", "EUR" of currency	Name of ISO Currency code
Bucket	<integer> one of {1,2}[13]	<integer> one of {1,2}	<integer> one of {1,2,3,4,5}[14]	<integer> one of {1,2}	Integer code relating to liquidity category of respective currencies
Label1	<string/float> one of GIRR_Tenors	<string/float> one of GIRR_Tenors	<string/float> one of GIRR_Tenors	<string/float> one of GIRR_Tenors	Tenor
Label2	<string> e.g. "Libor3m", "OIS"	<string> e.g. "Libor3m", "OIS"	<string> e.g. "Libor3m", "OIS"	<string> e.g. "Libor3m", "OIS"	Name of the curve
Amount	<float>	<float>	<float>	<float>	Delta sensitivity

Table 21: Field Guidance - GIRR Delta

GIRR_Tenors = {0.25, 0.5, 1, 2, 3, 5, 10, 15, 20, 30, "INFL", "XCCY"}

[12] Currently, the supported jurisdictions/rule-sets are Basel, CRR2, UK-PRA and US-NPR.

[13] 1 corresponds to illiquid and 2 to liquid i.e. divide the risk-weight by the $\sqrt{2}$

[14] 3,4 and 5 are relevant to CRR2 and the risk-weights that apply to the ERMII currencies, ref Article 325av (2)(3)(5).

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8.2 GIRR_VEGA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	Name of Variant
Sensitivity ID	<string> e.g. "S_IRV_xx"	<string> e.g. "S_IRV_xx"	<string> e.g. "S_IRV_xx"	<string> e.g. "S_IRV_xx"	Name of the sensitivity
RiskType	<string> "GIRR_VEGA"	<string> "GIRR_VEGA"	<string> "GIRR_VEGA"	<string> "GIRR_VEGA"	Name of Risk Type
Qualifier	<string> i.e. "USD", "EUR" of currency	<string> i.e. "USD", "EUR" of currency	<string> i.e. "USD", "EUR" of currency	<string> i.e. "USD", "EUR" of currency	Name of ISO Currency code
Label1	<float> one of GIRR_option_tenors	<float> one of GIRR_option_tenors	<float> one of GIRR_option_tenors	<float> one of GIRR_option_tenors	Option maturity Tenor
Label2	<float> one of GIRR_option_tenors	<float> one of GIRR_option_tenors	<float> one of GIRR_option_tenors	<float> one of GIRR_option_tenors	Underlying maturity tenor
Amount	<float>	<float>	<float>	<float>	Vega sensitivity for Variant 1 or vega for Variant 2
Label3	Variant 1: <empty> Variant2 :<float> e.g. 0.15	Variant 1: <empty> Variant2 :<float> e.g. 0.15	Variant 1: <empty> Variant2 :<float> e.g. 0.15	Variant 1: <empty> Variant2 :<float> e.g. 0.15	Denotes the implied volatility of the option (when Variant2 is used) otherwise empty

Table 22: Field Guidance - GIRR Vega

GIRR_Tenors = {0.25, 0.5, 1, 2, 3, 5, 10, 15, 20, 30, "INFL", "XCCY"}

GIRR_option_tenors = {0.5, 1, 3, 5, 10}

8.3 GIRR_CURV

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {"Variant 1a", "Variant 2a"}	<string> one of {"Variant 1a", "Variant 2a"}	<string> one of {"Variant 1a", "Variant 2a"}	<string> one of {"Variant 1a", "Variant 2a"}	Name of Variant
Sensitivity ID	<string> e.g. "S_IRC_xx"	<string> e.g. "S_IRC_xx"	<string> e.g. "S_IRC_xx"	<string> e.g. "S_IRC_xx"	Name of the sensitivity>. Note that curvature variants come in 2 or 3 rows. Need to be the same among these rows.
RiskType	<string> "GIRR_CURV"	<string> "GIRR_CURV"	<string> "GIRR_CURV"	<string> "GIRR_CURV"	Name of Risk Type
Qualifier	<string> e.g. "USD", "EUR" of currency	<string> e.g. "USD", "EUR" of currency	<string> e.g. "USD", "EUR" of currency	<string> e.g. "USD", "EUR" of currency	Name of ISO Currency code
Bucket	<integer> one of {1,2}	<integer> one of {1,2}	<integer> one of {1,2,3,4,5}	<integer> one of {1,2}	Integer code relating to liquidity category of respective currencies
Label1	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	Rate shift for calculating CVR

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Amount	<float>	<float>	<float>	<float>	Depends on variant. See section 6.2

Table 23: Field Guidance - GIRR Curvature

8.4 CSR_NS _DELTA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<empty>	<empty>	<empty>	<empty>	Not used for Delta risk types
Sensitivity ID	<string> e.g. S_CND_xx	<string> e.g. S_CND_xx	<string> e.g. S_CND_xx	<string> e.g. S_CND_xx	Name of the sensitivity
RiskType	<string> "CSR_NS_DELTA"	string> "CSR_NS_DELTA"	string> "CSR_NS_DELTA"	string> "CSR_NS_DELTA"	Name of Risk Type
Qualifier	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	Name of Issuer or Index name/ID
Bucket	<integer>	<integer>	<integer>	<integer>	Credit Spread – Non-Securitisation bucket
Label1	<integer/float> one of CSR_tenor	<integer/float> one of CSR_tenor	<integer/float> one of CSR_tenor	<integer/float> one of CSR_tenor	Tenor (year fraction)
Label2	<string> e.g "BOND", "CDS"	<string> e.g "BOND", "CDS"	<string> e.g "BOND", "CDS"	<string> e.g "BOND", "CDS"	Name of the Relevant issuer credit spread curve
Amount	<float>	<float>	<float>	<float>	Delta sensitivity
CreditQuality	<string> e.g. "AAA", "AA", "BBB"	<string> e.g. "IG", "SG", "SSG"	<string> e.g. "CQS1", "CQS2"	<string> e.g. "AAA", "AA", "BBB"	Issuer's credit quality

Table 24: Field Guidance - CSR_NS Delta

CSR_tenor = {0.5, 1, 3, 5 ,10}

8.5 CSR_NS _VEGA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	Name of Variant
Sensitivity ID	<string> e.g. "S_CNV_xx"	<string> e.g. "S_CNV_xx"	<string> e.g. "S_CNV_xx"	<string> e.g. "S_CNV_xx"	Name of the sensitivity
RiskType	<string> "CSR_NS_VEGA"	<string> "CSR_NS_VEGA"	<string> "CSR_NS_VEGA"	<string> "CSR_NS_VEGA"	Name of Risk Type
Qualifier	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	Name of Issuer or Index name/ID
Bucket	<integer>	<integer>	<integer>	<integer>	Credit Spread – Non-Securitisation bucket
Label1	<integer>	<integer>	<integer>	<integer>	Option Tenor

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Amount	<float>	<float>	<float>	<float>	Vega sensitivity for Variant 1 or vega for Variant 2
Label3	Variant 1: <empty> Variant2 :<float> e.g. 0.15	Variant 1: <empty> Variant2 :<float> e.g. 0.15	Variant 1: <empty> Variant2 :<float> e.g. 0.15	Variant 1: <empty> Variant2 :<float> e.g. 0.15	Denotes the implied volatility of the option (when Variant2 is used) otherwise empty
CreditQuality	<string> e.g. “AAA”, ”AA”, ”BBB”	<string> e.g. “IG”, “SG”, “SSG”	<string> e.g. “CQS1”, ”CQS2”	<string> e.g. “AAA”, ”AA”, ”BBB”	Issuer’s credit quality

Table 25: Field Guidance - CSR_NS Vega

8.6 CSR_NS _CURV

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {“Variant 1a”, “Variant 2a”}	<string> one of {“Variant 1a”, “Variant 2a”}	<string> one of {“Variant 1a”, “Variant 2a”}	<string> one of {“Variant 1a”, “Variant 2a”}	Name of Variant
Sensitivity ID	<string> e.g. “S_CNC_xx”	<string> e.g. “S_CNC_xx”	<string> e.g. “S_CNC_xx”	<string> e.g. “S_CNC_xx”	Name of the sensitivity
RiskType	<string> “CSR_NS_CURV”	<string> “CSR_NS_CURV”	<string> “CSR_NS_CURV”	<string> “CSR_NS_CURV”	Name of Risk Type
Qualifier	<string> e.g. “ISSUER xx”	<string> e.g. “ISSUER xx”	<string> e.g. “ISSUER xx”	<string> e.g. “ISSUER xx”	Name of Issuer or Index name/ID
Bucket	<integer>	<integer>	<integer>	<integer>	Credit Spread – Non-Securitisation bucket
Label1	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	Rate shift for calculating CVR
Amount	<float>	<float>	<float>	<float>	Depends on variant. See section 6.2
CreditQuality	<string> e.g. “AAA”, ”AA”, ”BBB”	<string> e.g. “IG”, “SG”, “SSG”	<string> e.g. “CQS1”, ”CQS2”	<string> e.g. “AAA”, ”AA”, ”BBB”	Issuer’s credit quality

Table 26: Field guidance - CSR_NS Curvature

8.7 CSR_SNC _DELTA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<empty>	<empty>	<empty>	<empty>	Not used for Delta risk types
Sensitivity ID	<string> e.g. “S_CSD_xx”	<string> e.g. “S_CSD_xx”	<string> e.g. “S_CSD_xx”	<string> e.g. “S_CSD_xx”	Name of the sensitivity
RiskType	<string> “CSR_SNC_DELTA”	<string> “CSR_SNC_DELTA”	<string> “CSR_SNC_DELTA”	<string> “CSR_SNC_DELTA”	Name of Risk Type
Qualifier	<string> e.g. “ISIN xx”	<string> e.g. “ISIN xx”	<string> e.g. “ISIN xx”	<string> e.g. “ISIN xx”	Name of Issuer, Tranche or Index name/ID
Bucket	<integer>	<integer>	<integer>	<integer>	Credit Spread – Securitisation non CTP bucket

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Label1	<string/float>	<string/float>	<string/float>	<string/float>	Tenor (year fraction)
Label2	<string> e.g. “BOND”, “CDS”	<string> e.g. “BOND”, “CDS”	<string> e.g. “BOND”, “CDS”	<string> e.g. “BOND”, “CDS”	Name of the Relevant issuer credit spread curve
CreditQuality	<string> e.g. “AAA”, ”AA”, ”BBB”	<string> e.g. “IG”, “SG”, “SSG”	<string> e.g. “CQS1”, ”CQS2”	<string> e.g. “AAA”, ”AA”, ”BBB”	Issuer’s credit quality
Amount	<float>	<float>	<float>	<float>	Amount of Delta Sensitivity

Table 27: Field Guidance - CSR_SNC Delta

8.8 CSR_SNC _VEGA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {“Variant 1”, “Variant 2”}	<string> one of {Variant 1, Variant 2}	<string> one of {Variant 1, Variant 2}	<string> one of {Variant 1, Variant 2}	Name of Variant
Sensitivity ID	<string> e.g. “S_CSV_xx”	<string> e.g. “S_CSV_xx”	<string> e.g. “S_CSV_xx”	<string> e.g. “S_CSV_xx”	Name of the sensitivity
RiskType	<string> “CSR_SNC_VEGA”	<string> CSR_SNC_VEGA	<string> CSR_SNC_VEGA	<string> CSR_SNC_VEGA	Name of Risk Type
Qualifier	<string> e.g. “ISIN xx”	<string> ie ISIN xx	<string> ie ISIN xx	<string> ie ISIN xx	Name of Issuer or Index name/ID
Bucket	<integer>	<integer>	<integer>	<integer>	Credit Spread – Securitisation non-CTP bucket
Label1	<integer>	<integer>	<integer>	<integer>	Option Tenor
Amount	<float>	<float>	<float>	<float>	Vega sensitivity for Variant 1 or vega for Variant 2
Label3	<float> e.g. 0.15	<float> ie 0.15	<float> ie 0.15	<float> ie 0.15	Denotes the implied volatility of the option (when Variant2 is used)
CreditQuality	<string> e.g. “AAA”, ”AA”, ”BBB”	<string> e.g. “IG”, “SG”, “SSG”	<string> e.g. “CQS1”, ”CQS2”	<string> e.g. “AAA”, ”AA”, ”BBB”	Issuer’s credit quality

Table 28: Field Guidance - CSR_SNC Vega

8.9 CSR_SNC _CURV

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {“Variant 1a”, “Variant 2a”}	<string> one of {“Variant 1a”, “Variant 2a”}	<string> one of {“Variant 1a”, “Variant 2a”}	<string> one of {“Variant 1a”, “Variant 2a”}	Name of Variant
Sensitivity ID	<string> e.g. “S_CSC_xx”	<string> e.g. “S_CSC_xx”	<string> e.g. “S_CSC_xx”	<string> e.g. “S_CSC_xx”	Name of the sensitivity
RiskType	<string> “CSR_SNC_CURV”	<string> “CSR_SNC_CURV”	<string> “CSR_SNC_CURV”	<string> “CSR_SNC_CURV”	Name of Risk Type

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Qualifier	<string> e.g “ISSUER xx”	<string> e.g “ISSUER xx”	<string> e.g “ISSUER xx”	<string> e.g “ISSUER xx”	Name of Issuer or Index name/ID
Bucket	<integer>	<integer>	<integer>	<integer>	Credit Spread – Securitisation non-CTP bucket
Label1	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	Rate shift for calculating CVR
Amount	<float>	<float>	<float>	<float>	Depends on variant. See section 6.2
CreditQuality	<string> e.g. “AAA”, ”AA”, ”BBB”	<string> e.g. “IG”, “SG”, “SSG”	<string> e.g. “CQS1”, ”CQS2”	<string> e.g. “AAA”, ”AA”, ”BBB”	Issuer’s credit quality

Table 29: Field Guidance - CSR_SNC Curvature

8.10 CSR_SC _DELTA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<empty>	<empty>	<empty>	<empty>	Not used for Delta risk types
Sensitivity ID	<string> e.g. “S_CCD_xx”	<string> e.g. “S_CCD_xx”	<string> e.g. “S_CCD_xx”	<string> e.g. “S_CCD_xx”	Name of the sensitivity
RiskType	<string> “CSR_SC_DELTA ”	<string> “CSR_SC_DELTA ”	<string> “CSR_SC_DELTA ”	<string> “CSR_SC_DELTA ”	Name of Risk Type
Qualifier	<string> e.g. “ISSUER xx”	<string> e.g. “ISSUER xx”	<string> e.g. “ISSUER xx”	<string> e.g. “ISSUER xx”	Name of Issuer, Tranche or Index name/ID
Bucket	<integer>	<integer>	<integer>	<integer>	Credit Spread – Securitisation CTP bucket
Label1	<string/float>	<string/float>	<string/float>	<string/float>	Tenor (year fraction)
Label2	<string> e.g. “BOND”, “CDS”	<string> e.g. “BOND”, “CDS”	<string> e.g. “BOND”, “CDS”	<string> e.g. “BOND”, “CDS”	Name of the Relevant issuer credit spread curve
CreditQuality	<string> e.g. “AAA”, ”AA”, ”BBB”	<string> e.g. “IG”, “SG”, “SSG”	<string> e.g. “CQS1”, ”CQS2”	<string> e.g. “AAA”, ”AA”, ”BBB”	Issuer’s credit quality
Amount	<float>	<float>	<float>	<float>	Amount of Delta Sensitivity

Table 30: Field Guidance - CSR_SC Delta

8.11 CSR_SC _VEGA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {“Variant 1”, “Variant 2”}	<string> one of {“Variant 1”, “Variant 2”}	<string> one of {“Variant 1”, “Variant 2”}	<string> one of {“Variant 1”, “Variant 2”}	Name of Variant

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Sensitivity ID	<string> e.g. "S_CCV_xx"	<string> e.g. "S_CCV_xx"	<string> e.g. "S_CCV_xx"	<string> e.g. "S_CCV_xx"	Name of the sensitivity
RiskType	<string> "CSR_SC_VEGA"	<string> "CSR_SC_VEGA"	<string> "CSR_SC_VEGA"	<string> "CSR_SC_VEGA"	Name of Risk Type
Qualifier	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	Name of Issuer or Index name/ID
Bucket	<integer>	<integer>	<integer>	<integer>	Credit Spread – Securitisation CTP bucket
Label1	<integer>	<integer>	<integer>	<integer>	Option Tenor
Amount	<float>	<float>	<float>	<float>	Vega sensitivity for Variant 1 or vega for Variant 2
Label3	<float> e.g. 0.15	<float> e.g. 0.15	<float> e.g. 0.15	<float> e.g. 0.15	Denotes the implied volatility of the option (when Variant2 is used)
CreditQuality	<string> e.g. "AAA", "AA", "BBB"	<string> e.g. "IG", "SG", "SSG"	<string> e.g. "CQS1", "CQS2"	<string> e.g. "AAA", "AA", "BBB"	Issuer's credit quality

Table 31: Field Guidance - CSR_SC Vega

8.12 CSR_SC _CURV

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {"Variant 1a", "Variant 2a"}	<string> one of {"Variant 1a", "Variant 2a"}	<string> one of {"Variant 1a", "Variant 2a"}	<string> one of {"Variant 1a", "Variant 2a"}	Name of Variant
Sensitivity ID	<string> e.g. "S_CCC_xx"	<string> e.g. "S_CCC_xx"	<string> e.g. "S_CCC_xx"	<string> e.g. "S_CCC_xx"	Name of the sensitivity
RiskType	<string> "CSR_SC_CURV"	<string> "CSR_SC_CURV"	<string> "CSR_SC_CURV"	<string> "CSR_SC_CURV"	Name of Risk Type
Qualifier	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	Name of Issuer or Index name/ID
Bucket	<integer>	<integer>	<integer>	<integer>	Credit Spread – Securitisation CTP bucket
Label1	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	Rate shift for calculating CVR
CreditQuality	<string> e.g. "AAA", "AA", "BBB"	<string> e.g. "IG", "SG", "SSG"	<string> e.g. "CQS1", "CQS2"	<string> e.g. "AAA", "AA", "BBB"	Issuer's credit quality
Amount	<float>	<float>	<float>	<float>	Depends on variant. See section 6.2

Table 32: Field Guidance - CSR_SC Curvature

8.13 COMM _DELTA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description

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Variant	<empty>	<empty>	<empty>	<empty>	Not used for Delta risk types
Sensitivity ID	<string> e.g. "S_CMD_xx"	<string> e.g. "S_CMD_xx"	<string> e.g. "S_CMD_xx"	<string> e.g. "S_CMD_xx"	Name of the sensitivity
RiskType	<string> "COMM_DELTA"	<string> "COMM_DELTA"	<string> "COMM_DELTA"	<string> "COMM_DELTA"	Name of Risk Type
Qualifier	<string> e.g. "COAL"," WTI"	<string> e.g. "COAL"," WTI"	<string> e.g. "COAL"," WTI"	<string> e.g. "COAL"," WTI"	Name of Commodity name/ID
Bucket	<integer> e.g. 3	<integer> e.g. 3	<integer> e.g. 3	<integer/float/string> e.g. 3, 3a	Commodity bucket
Label1	<integer/float> one of {0, 0.25, 0.5, 1,2,…}	<integer/float> one of {0, 0.25, 0.5, 1,2,…}	<integer/float> one of {0, 0.25, 0.5, 1,2,…}	<integer/float> one of {0, 0.25, 0.5, 1,2,…}	Tenor (year fraction)
Label2	<string> e.g. "LONDON", "TEXAS"	<string> e.g. "LONDON", "TEXAS"	<string> e.g. "LONDON", "TEXAS"	<string> e.g. "LONDON", "TEXAS"	Name of the Commodity delivery location
Amount	<float>	<float>	<float>	<float>	Amount of Delta Sensitivity

Table 33: Field Guidance - Commodity Delta

8.14 COMM _VEGA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {"Variant 1", "Variant 2"}}	<string> one of {"Variant 1", "Variant 2"}}	<string> one of {"Variant 1", "Variant 2"}}	<string> one of {"Variant 1", "Variant 2"}}	Name of Variant
Sensitivity ID	<string> e.g. "S_CMV_xx"	<string> e.g. "S_CMV_xx"	<string> e.g. "S_CMV_xx"	<string> e.g. "S_CMV_xx"	Name of the sensitivity
RiskType	<string> "COMM_VEGA"	<string> "COMM_VEGA"	<string> "COMM_VEGA"	<string> "COMM_VEGA"	Name of Risk Type
Qualifier	<string> e.g. "COAL", "WTI"	<string> e.g. "COAL", "WTI"	<string> e.g. "COAL", "WTI"	<string> e.g. "COAL", "WTI"	Name of Commodity name/ID
Bucket	<integer> e.g. 3	<integer> e.g. 3	<integer> e.g. 3	<integer/float> ie 3, 3a	Commodity bucket
Label1	<integer/float> one of {0.25, 0.5, 1,2,…}	<integer/float> one of {0.25, 0.5, 1,2,…}	<integer/float> one of {0.25, 0.5, 1,2,…}	<integer/float> one of {0.25, 0.5, 1,2,…}	Option Tenor (year fraction)
Amount	<float>	<float>	<float>	<float>	Vega sensitivity for Variant 1 or vega for Variant 2
Label3	<float> e.g. 0.15	<float> e.g. 0.15	<float> e.g. 0.15	<float> e.g. 0.15	Denotes the implied volatility of the option (when Variant2 is used)

Table 34: Field Guidance - Commodity Vega

8.15 COMM _CURV

Field Name	Basel	US-NPR	CRR	UK_PRA	Description

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Variant	<string> one of {"Variant 1a", "Variant 2a"}}	<string> one of {"Variant 1a", "Variant 2a"}}	<string> one of {"Variant 1a", "Variant 2a"}}	<string> one of {"Variant 1a", "Variant 2a"}}	Name of Variant
Sensitivity ID	<string> e.g.. "S_CMC_xx"	<string> e.g.. "S_CMC_xx"	<string> e.g.. "S_CMC_xx"	<string> e.g.. "S_CMC_xx"	Name of the sensitivity
RiskType	<string> "COMM_CURV"	<string> COMM_CURV	<string> COMM_CURV	<string> COMM_CURV	Name of Risk Type
Qualifier	<string> e.g. "COAL", "WTI"	<string> e.g. "COAL", "WTI"	<string> e.g. "COAL", "WTI"	<string> e.g. "COAL", "WTI"	Name of Commodity name/ID
Bucket	<integer> e.g. 3	<integer> e.g. 3	<integer> e.g. 3	<integer/float> ie 3, 3a	Commodity bucket
Label1	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	Rate shift for calculating CVR
Amount	<float>	<float>	<float>	<float>	Depends on variant. See section 6.2

Table 35: Field Guidance - Commodity Curvature

8.16 EQ_DELTA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<empty>	<empty>	<empty>	<empty>	Not used for Delta risk types
Sensitivity ID	<string> e.g. "S_EQD_xx"	<string> e.g. "S_EQD_xx"	<string> e.g. "S_EQD_xx"	<string> e.g. "S_EQD_xx"	Name of the sensitivity
RiskType	<string> "EQ_DELTA"	<string> "EQ_DELTA"	<string> "EQ_DELTA"	<string> "EQ_DELTA"	Name of Risk Type
Qualifier	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	Name of Issuer, Index name/ID
Bucket	<integer> e.g. 2	<integer> e.g. 2	<integer> e.g. 2	<integer> e.g. 2	Equity bucket
Label2	<string> one of {"SPOT", "REPO"}	<string> one of {"SPOT", "REPO"}	<string> one of {"SPOT", "REPO"}	<string> one of {"SPOT", "REPO"}	Denotes whether the sensitivity is against a "SPOT" equity risk factor or against an equity "REPO" rate.
Amount	<float>	<float>	<float>	<float>	Amount of Delta Sensitivity

Table 36: Field Guidance - Equity Delta

8.17 EQ _VEGA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description

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Variant	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	Name of Variant
Sensitivity ID	<string> e.g. "S_CMV_xx"	<string> e.g. "S_CMV_xx"	<string> e.g. "S_CMV_xx"	<string> e.g. "S_CMV_xx"	Name of the sensitivity
RiskType	<string> "EQ_VEGA"	<string> "EQ_VEGA"	<string> "EQ_VEGA"	<string> "EQ_VEGA"	Name of Risk Type
Qualifier	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	Name of Issuer, Index name/ID
Bucket	<integer> e.g. 2	<integer> e.g. 2	<integer> e.g. 2	<integer> e.g. 2	Equity bucket
Label1	<integer> e.g. 5	<integer> e.g. 5	<integer> e.g. 5	<integer> e.g. 5	Option Tenor
Amount	<float>	<float>	<float>	<float>	Vega sensitivity for Variant 1 or vega for Variant 2
Label3	<float> e.g. 0.15	<float> e.g. 0.15	<float> e.g. 0.15	<float> e.g. 0.15	Denotes the implied volatility of the option (when Variant2 is used)

Table 37: Field Guidance - Equity Vega

8.18 EQ _CURV

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {"Variant 1a", "Variant 2a"}}	<string> one of {"Variant 1a", "Variant 2a"}}	<string> one of {"Variant 1a", "Variant 2a"}}	<string> one of {"Variant 1a", "Variant 2a"}}	Name of Variant
Sensitivity ID	<string> e.g. "S_EQC_xx"	<string> e.g. "S_EQC_xx"	<string> e.g. "S_EQC_xx"	<string> e.g. "S_EQC_xx"	Name of the sensitivity
RiskType	<string> "EQ_CURV"	<string> "EQ_CURV"	<string> "EQ_CURV"	<string> "EQ_CURV"	Name of Risk Type
Qualifier	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	Name of Issuer, Index name/ID
Bucket	<integer> e.g. 2	<integer> e.g. 2	<integer> e.g. 2	<integer> e.g. 2	Equity bucket
Label1	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	Rate shift for calculating CVR
Amount	<float>	<float>	<float>	<float>	Depends on variant. See section 6.2

Table 38: Field Guidance - Equity Curvature

8.19 FX _DELTA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description

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Variant	<empty>	<empty>	<empty>	<empty>	Not used for Delta risk types
Sensitivity ID	<string> e.g "S_FXD_xx"	<string> e.g "S_FXD_xx"	<string> e.g "S_FXD_xx"	<string> e.g "S_FXD_xx"	Name of the sensitivity
RiskType	<string> "FX_DELTA"	<string> "FX_DELTA"	<string> "FX_DELTA"	<string> "FX_DELTA"	Name of Risk Type
Qualifier	<string> e.g., "USD", "EUR"	<string> e.g., "USD", "EUR"	<string> e.g., "USD", "EUR"	<string> e.g., "USD", "EUR"	Name of ISO Currency code
Bucket	<integer> one of {1,2}[15]	<integer> one of {1,2}	<integer> one of {1,2,3,4,5}[16]	<integer> one of {1,2}	Integer code relating to liquidity category of respective currencies
Amount	<float>	<float>	<float>	<float>	Amount of Delta Sensitivity

Table 39: Field Guidance - FX Delta

8.20 FX _VEGA

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	Name of Variant
Sensitivity ID	<string> e.g. "S_FXV_xx"	<string> e.g. "S_FXV_xx"	<string> e.g. "S_FXV_xx"	<string> e.g. "S_FXV_xx"	Name of the sensitivity
RiskType	<string> "FX_VEGA"	<string> "FX_VEGA"	<string> "FX_VEGA"	<string> "FX_VEGA"	Name of Risk Type
Qualifier	<string> e.g., "EURUSD", "EURGBP"	<string> e.g., "EURUSD", "EURGBP"	<string> e.g., "EURUSD", "EURGBP"	<string> e.g., "EURUSD", "EURGBP"	Currency pair
Label1	<integer/float> one from FX_Option_Tenors	<integer/float> one from FX_Option_Tenors	<integer/float> one from FX_Option_Tenors	<integer/float> one from FX_Option_Tenors	Option tenor
Amount	<float>	<float>	<float>	<float>	Vega sensitivity for Variant 1 or vega for Variant 2
Label3	<float> e.g. 0.15	<float> e.g. 0.15	<float> e.g. 0.15	<float> e.g. 0.15	Denotes the implied volatility of the option (when Variant2 is used)

Table 40: Field Guidance - FX Vega

FX_Option_Tenors = {0.5, 1, 3, 5, 10}

8.21 FX _CURV

Field Name	Basel	US-NPR	CRR	UK_PRA	Description

[15] 1 corresponds to illiquid and 2 to liquid i.e. divide the risk-weight by the $\sqrt{2}$

[16] 3,4 and 5 are relevant to CRR2 and the risk-weights that apply to the ERMII currencies, ref Article 325av (2)(3)(5).

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Variant	<string> one of {"Variant 1a", "Variant 2a"}	<string> one of {"Variant 1a", "Variant 2a"}	<string> one of {"Variant 1a", "Variant 2a"}	<string> one of {"Variant 1a", "Variant 2a"}	Name of Variant
Sensitivity ID	<string> e.g. "S_FXC_xx"	<string> e.g. "S_FXC_xx"	<string> e.g. "S_FXC_xx"	<string> e.g. "S_FXC_xx"	Name of the sensitivity
RiskType	<string> "FX_CURV"	<string> "FX_CURV"	<string> "FX_CURV"	<string> "FX_CURV"	Name of Risk Type
Qualifier	<string> e.g., "USD", "EUR"	<string> e.g., "USD", "EUR"	<string> e.g., "USD", "EUR"	<string> e.g., "USD", "EUR"	Name of ISO Currency code
Bucket	<integer> one of {1,2}[17]	<integer> one of {1,2}	<integer> one of {1,2,3,4,5}[18]	<integer> one of {1,2}	Integer code relating to liquidity category of respective currencies
Label1	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	<float> e.g. 0.001	Rate shift for calculating CVR
Amount	<float>	<float>	<float>	<float>	Depends on variant. See section 6.2

Table 41: Field Guidance - FX Curvature

8.22 DRC_NS

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {"Variant 1", "Variant 2", "Variant 3"}	<string> one of {"Variant 1", "Variant 2", "Variant 3"}	<string> one of {"Variant 1", "Variant 2", "Variant 3"}	<string> one of {"Variant 1", "Variant 2", "Variant 3"}	Name of Variant
Sensitivity ID	<string> e.g. "S_DNS_xx"	<string> e.g. "S_DNS_xx"	<string> e.g. "S_DNS_xx"	<string> e.g. "S_DNS_xx"	Name of the sensitivity
RiskType	<string> "DRC_NS"	<string> "DRC_NS"	<string> "DRC_NS"	<string> "DRC_NS"	Name of Risk Type
Qualifier	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	<string> e.g. "ISSUER xx"	Name of Issuer /ID
Bucket	<string> one of: {CORPORATES, SOVEREIGNS, LOCAL_GOV}	<string> one of: {CORPORATES, NON_US_SOVEREIGNS, PSE_GSE_DEBT, DEFAULTED}	As in Basel	As in Basel	DRC_NS bucket
Label2	<string> one of: {EQUITY, NON-SENIOR, SENIOR}	<string> one of: {EQUITY, NON-SENIOR, SENIOR, ZERO_LGD, GSE_GUARANTEED, GSE_NOT_GUARANTEED, DEFAULTED	As in Basel	As in Basel	Instrument seniority
Amount	<float>	<float>	<float>	<float>	Depends on variant. V1: Market Value, V2: JTD (unscaled), V3: maturity scaled.

[17] 1 corresponds to illiquid and 2 to liquid i.e. divide the risk-weight by the $\sqrt{2}$

[18] 3,4 and 5 are relevant to CRR2 and the risk-weights that apply to the ERMII currencies, ref Article 325av (2)(3)(5).

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Label3	<empty> or <float> e.g. 1,200,000.00	<empty> or <float> e.g. 1,200,000.00	<empty> or <float> e.g. 1,200,000.00	<empty> or <float> e.g. 1,200,000.00	V1: Notional. V2 and V3: <empty>
EndDate	<empty> or <string> e.g. “2019-08-15”	<empty> or <string> or “2019-08-15”	<empty> or <string> e.g. “2019-08-15”	<empty> or <string> e.g. “2019-08-15”	Maturity date ISO-Standard "YYYY-MM-DD". V3: <empty>
CreditQuality	<string> one of: {AAA, AA, A, BBB, BB, B, CCC, CC, C, UNRATED, DEFAULTED, ZERO_RW}	<string> one of: {IG, SG, SSG, DEFAULTED}	<string> one of: {CQS1, CQS2, CQS3, CQS4, CQS5, CQS6, UNRATED, DEFAULTED, ZERO_RW}	As in Basel	Issuer’s credit quality
LongShortInd	<string> one of {“L”,”S”}	<string> one of {“L”,”S”}	<string> one of {“L”,”S”}	<string> one of {“L”,”S”}	“L” (long) or “S” (short) indicator
CoveredBondInd	<boolean> one of {“Y”,”N”}	<boolean> one of {“Y”,”N”}	<boolean> one of {“Y”,”N”}	<boolean> one of {“Y”,”N”}	Covered Bond indicator

Table 42: Field Guidance - DRC_NS

8.23 DRC_SNC

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {“Variant 1”, “Variant 2”}	<string> one of {“Variant 1”, “Variant 2”}	<string> one of {“Variant 1”, “Variant 2”}	<string> one of {“Variant 1”, “Variant 2”}	Name of Variant
Sensitivity ID	<string> e.g. “S_DNC_xx”	<string> e.g. “S_DNC_xx”	<string> e.g. “S_DNC_xx”	<string> e.g. “S_DNC_xx”	Name of the sensitivity
RiskType	<string> “DRC_SNC”	<string> “DRC_SNC”	<string> “DRC_SNC”	<string> “DRC_SNC”	Name of Risk Type
Qualifier	<string> e.g. “ISIN xx”	<string> e.g. “ISIN xx”	<string> e.g. “ISIN xx”	<string> e.g. “ISIN xx”	Name of Issuer /ID
Bucket	<string> e.g. “CORPORATES” or “RMBS_EUROPE”	<string> e.g. “CORPORATES” or “RMBS_EUROPE”	<string> e.g. “CORPORATES” or “RMBS_EUROPE”	<string> e.g. “CORPORATES” or “RMBS_EUROPE”	DRC_SNC bucket
Label2	V1: <string> “NON-SENIOR”, “SENIOR” V2: <empty>	V1: Not applicable V2: <empty>	V1: <string> “NON-SENIOR”, “SENIOR” V2: <empty>	V1: <string> “NON-SENIOR”, “SENIOR” V2: <empty>	Instrument seniority for Variant 1, <empty> for Variant 2
Amount	<float>	<float>	<float>	<float>	JTD amount which is equivalent to market value.
EndDate	<string> ie “2019-08-15”	<string> ie “2019-08-15”	<string> ie “2019-08-15”	<string> ie “2019-08-15”	Maturity date or option expiry in ISO-Standard "YYYY-MM-DD"
CreditQuality	V1: <string> e.g. “AAA”, ”AA”, ”BBB” V2: <float>	V2: <float>	V1: <string> e.g. “CQS1”, ”CQS2” V2: <float>	V1: <string> e.g. “AAA”, ”AA”, ”BBB” V2: <float>	V1: Credit rating V2: risk-weight
LongShortInd	<string> one of {“L”,”S”}	<string> one of {“L”,”S”}	<string> one of {“L”,”S”}	<string> one of {“L”,”S”}	“L” (long) or “S” (short) indicator
TrancheThicknes s	<empty> or <float> between [0,1] e.g. 0.3	<empty>	<empty> or <float> between [0,1] e.g. 0.3	<empty> or <float> between [0,1] e.g. 0.3	V1: Tranche thickness V2: <empty>

Table 43: Field Guidance - DRC_SNC

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Variant 1 corresponds to SEC-ERBA approach for calculating risk-weights. Variant 1 does not apply to US-NPR jurisdiction. Variant 2 to risk-weights as an input. See CreditQuality column.

8.24 DRC_SC

Field Name	Basel	US-NPR	CRR	UK_PRA	Description
Variant	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	<string> one of {"Variant 1", "Variant 2"}	Name of Variant
Sensitivity ID	<string> e.g. "S_DCD_xx"	<string> e.g. "S_DCD_xx"	<string> e.g. "S_DCD_xx"	<string> e.g. "S_DCD_xx"	Name of the sensitivity
RiskType	<string> "DRC_SC"	<string> "DRC_SC"	<string> "DRC_SC"	<string> "DRC_SC"	Name of Risk Type
Qualifier	<string> e.g. "SERIES xx"	<string> e.g. "SERIES xx"	<string> e.g. "SERIES xx"	<string> e.g. "SERIES xx"	Name of Series of the index or "Bespoke portfolio"
Bucket	<string> e.g. "ITRAXX IG"	<string> e.g. "ITRAXX IG"	<string> e.g. "ITRAXX IG"	<string> e.g. "ITRAXX IG"	DRC_SC bucket
Label1	<string> e.g."4-125"	<string> e.g."4-125"	<string> e.g."4-125"	<string> e.g."4-125"	Names in the tranche.
Label2	<string> one of: {TRANCHED_INDEX, NON_TRANCHED_INDEX, SINGLE_NAME_HEDGE, BESPOKE_EXPOSURE}	<string> one of: {TRANCHED_INDEX, NON_TRANCHED_INDEX, SINGLE_NAME_HEDGE, BESPOKE_EXPOSURE}	<string> one of: {TRANCHED_INDEX, NON_TRANCHED_INDEX, SINGLE_NAME_HEDGE, BESPOKE_EXPOSURE}	<string> one of: {TRANCHED_INDEX, NON_TRANCHED_INDEX, SINGLE_NAME_HEDGE, BESPOKE_EXPOSURE}	Categorisation of modelled exposure.
Amount	<float>	<float>	<float>	<float>	JTD/Market value
EndDate	<string> ie "2019-08-15"	<string> ie "2019-08-15"	<string> ie "2019-08-15"	<string> ie "2019-08-15"	Maturity date or option expiry in ISO-Standard "YYYY-MM-DD"
CreditQuality	V2: <string> e.g. "AAA", "AA", "BBB" V1: <float>	V2: <string> e.g. "IG", "SG" V1: <float>	V2: <string> e.g. "CQS1", "CQS2" V1: <float>	V2: <string> e.g. "AAA", "AA", "BBB" V1: <float>	V1: risk-weight V2: Credit rating
LongShortInd	<string> one of {"L","S"}	<string> one of {"L","S"}	<string> one of {"L","S"}	<string> one of {"L","S"}	"L" (long) or "S" (short) indicator

Table 44: Field Guidance - DRC_SC

Variant 1 corresponds to banking book approach where the risk-weights are direct inputs in the CreditQuality column. Variant 2 corresponds to decomposition approach where the assumption is that we can source the risk-weights using the DRC Non-Securitisation rules.

8.25 RRAO

Field Name	Basel	US-NPR	CRR	UK_PRA	Description

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Variant	<empty>	<empty>	<empty>	<empty>	One variant – Gross notional amount provided as input
Sensitivity ID	<string> e.g. "S_RRAO_xx"	<string> e.g. "S_RRAO_xx"	<string> e.g. "S_RRAO_xx"	<string> e.g. "S_RRAO_xx"	Name of the sensitivity
RiskType	<string> one of ("RRAO_1_PERCENT ", "RRAO_01_PERCENT"}	<string> one of ("RRAO_1_PERCENT ", "RRAO_01_PERCENT"}	<string> one of ("RRAO_1_PERCENT ", "RRAO_01_PERCENT"}	<string> one of ("RRAO_1_PERCENT ", "RRAO_01_PERCENT"}	Name of Risk Type
Amount	<float>	<float>	<float>	<float>	Notional amount of the position

Table 45: Field Guidance – RRAO

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9 CRIF files and ISDA ecosystem

9.1 The CRIF file standard in ISDA FRTB-SA calculator and Unit Test

ISDA has built an ecosystem of tools around the CRIF standard in order to enhance our solutions suite. We have developed the FRTB-SA Unit Test which has been validated by numerous markets participants and constitutes the industry's golden standard in regards to validating FRTB-SA implementations.

Furthermore, ISDA developed its own implementation of an FRTB-SA calculator which can be accessed through our online platform Perun©. Both these tools are built around the CRIF representation of FRTB-SA inputs and are compatible with it. For example, interested users can upload CRIF files in Perun and calculate FRTB-SA capital.

9.2 Controlling FRTB-SA calculations with configuration Flags

Our commitment to regulatory compliance and adaptability is reflected in our capability to interpret CRIF files across a spectrum of regulatory jurisdictions. We amplify this by integrating numerous configuration flags within the CRIF files.

These configuration flags align with the currently supported regulatory frameworks specifically:

- MAR from Basel Committee on Banking Supervision (BCBS)
- European Union's Capital Requirements Regulation (CRR2), and;
- UK's Prudential Regulation Authority (PRA) regulations.

The configuration flags act as dynamic parameters in risk calculations. Each flag represents an alternative interpretation or application of capital requirement rules, offering a nuanced approach to adjusting risk evaluations to match specific regulatory standards.

Specifically, the list of configuration flags includes:

9.2.1 CRR2_RW_INFL_XCCY

This configuration flag is associated with the interpretation of cross-currency (XCCY) and inflation (INFL) risk weights under the CRR2 regulations for liquid currencies. There are two alternatives:

Alt1: INFL and XCCY risk factor risk weights for liquid currencies are divided by $\sqrt{2}$.

Alt2: CRR2 Article 325ae (3) can be read so as to imply that GIRR DELTA XCCY and INFL are not risk-free tenors and hence the RW division by $\sqrt{2}$ does not apply to them.

9.2.2 VEGA_CORR_INFL_XCCY:

This flag presents alternatives for calculating the GIRR Vega intra-bucket correlation for inflation (INFL) and cross-currency (XCCY) risks.

Alt 1: Follows the rules as described in MAR21.93 and MAR21.94. Specifically for $\rho_{option_maturity}$:

- $\rho_{XCCY,INFL} = 0$
- $\rho_{XCCY,i} = 0$
- $\rho_{INFL,i} = 0.4$

Alt 2: the correlation $\rho_{option_maturity} = 1$ for all cases ($\rho_{XCCY,i} = 1$ and $\rho_{IFNL,i} = 1$).

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Alt3: the formula in MAR21.93 for $\rho_{option_maturity}$ is used but an underlying maturity of 0.5 years is used for the INFL and XCCY tenors (T_{INFL} = 0.5 and T_{XCCY} = 0.5).

Assumption 1 presents three alternatives. In the first one, correlations are determined based on the factors' characteristics, following MAR21.94. The second alternative sets a correlation of 1 for all combinations, while the third uses the formula in MAR21.93 with an underlying maturity of 0.5 years used in all cases.

9.2.3 CSR_NS_INDX_BUCKET_NAME_CORRELATION

Under Basel, as per MAR21.55 the name correlation applied within the index buckets 17 and 18 is set to be ρ_{name} = 80%.

However, under CRR2, as per Article 325ai the name correlation applied within the index buckets 19 and 20 is set to be the same as within any other buckets i.e. ρ_{name} = 35%.

The above gap has not been fixed by the EU regulator as part of the Jan 2020 DA. It appears that the regulator is aware but they didn't have the mandate to fix it. Therefore:

Alt 1: ρ_{name} = 35% for buckets 19 and 20 under CRR2. This is in line with the current CRR2 regulation.

Alt 2: ρ_{name} = 80% for buckets 19 and 20 under CRR2. This is in line with the expectation that the relevant gap between Basel and CRR2 will be bridged.

UK-PRA follows the Alt2 convention i.e. the applied ρ_{name} = 80%.

9.2.4 DRC_NS_COVERED_SENIORITY

This flag relates to how "covered" bonds are treated in default risk charge (DRC) calculations for non-securitisations. Alternative 1 adjusts the loss given default (LGD) of the exposure to 0.25, with the bond belonging to the seniority class specified under Label2. Alternative 2 treats "covered" bonds as a separate seniority class, ranked as EQUITY < NON-SENIOR < SENIOR < COVERED:

Alt 1: "Covered" bond entry (see CoveredBondInd column) overrides the LGD of the exposure with 0.25. The exposure belongs to the seniority class specified under Label2.

Alt2: "Covered" bond is considered as a separate seniority class with seniority level as EQUITY < NON-SENIOR < SENIOR < COVERED.

9.2.5 CRR2_CSR_NS_INDX_RATING_CORR:

This flag is applicable only to CRR2 ruleset and for Credit Spread Risk (CSR) Non-Securitisation (NS) Index Buckets.

The inter-bucket γ_{rating} correlation that is applicable for the two CSR NS index buckets, is a point of divergence between the Basel and CRR2 implementations.

Specifically Basel suggests that this correlation should be 1 while Article 325j of CRR2 and UK-PRA CP specify that it should be 0.5.

We have therefore introduced two alternatives for the CRR2 calculation:

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Alt 1: $\gamma_{rating} = 0.5$ thus $\gamma = (\gamma_{rating} * \gamma_{sector}) = 0.5 * 0.75 = 0.375$

Alt 2: $\gamma_{rating} = 1$ thus $\gamma = (\gamma_{rating} * \gamma_{sector}) = 1 * 0.75 = 0.75$

UK-PRA follows the Alt1 convention i.e. the applied $\gamma_{rating} = 0.5$.

10 References

Basel Committee on Banking Supervision (BCBS). (2019, December 15). *MAR - Calculation of RWA for market risk*. Retrieved from https://www.bis.org/basel_framework/standard/MAR.htm

Basel Committee on Banking Supervision (BCBS). (January 2019 (rev. February 2019)). *Minimum capital requirments for market risk.*

International Swaps Dealers Association (ISDA). (n.d.). *ISDA SIMM Methodology: Risk Data Standards.*

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